

VDB_Checklist

What this is: every VectoraDB feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `VDB_Feature_Implemented.pdf` (same ids).

Last updated: 17 September 2026 · **Reflects:** `main @ 56432a4` (v0.8.7, test isolation PR #41) + Blackbox precision (`fix/blackbox-precision`)

Source: `docs/VDB_Checklist.html` → `make feature-doc` → this PDF

☒ Done & tested — automated tests (unit, VM integration or CI)

☒ Done — tested by hand only

☐ Partial — gaps, or not verified end to end

☒ Known issue

☐ Not started

Sub-features: ☒ built ☐ partly ☐ not built yet ☒ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	·	·	78%
B · CLI & services	2	1	·	1	·	·	50%
C · Branching & storage	8	6	1	1	·	·	88%
D · Durability	4	2	·	2	·	·	50%
E · Security foundations	3	2	1	·	·	·	100%
F · SQL gateway	7	6	1	·	·	·	100%
G · Accounts & auth	5	4	·	1	·	·	80%
H · Blackbox (record of schema changes)	18	16	·	2	·	·	89%
I · Import & migration	13	4	8	1	·	·	92%
J · ETL pipelines (experimental)	4	2	2	·	·	·	100%
K · APIs, SDKs & agents	6	3	2	1	·	·	83%
L · Web console	10	·	9	1	·	·	90%
M · Build, CI & releases	6	4	2	·	·	·	100%
Total (features built or started)	95	54	29	12	·	·	87%
N · Not started (roadmap)	11	·	·	·	·	11	0%

Storage engine — not implemented, by design

VectoraDB does **not** have its own storage engine. Data is stored by **PostgreSQL 16's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:16-bookworm` image plus `wal-g`. This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Run `vdb update` against a real release on macOS and Windows (A9) — cut v0.9.0, install it, then update to v0.9.1.
- HA beyond the demo: archive WAL while main runs on the promoted standby, then automatic failover and fencing (D4). (*Safety fixes C3, D4 hazards, H7, I5, I6 done.*)
- Verify downloads everywhere: installers and `vdb setup`'s engine download still skip `SHA256SUMS` (A1, A2, A7).

4. Check the web pages that were built but never used signed in: Integrity, Policies, sidebar (L8–L10); add Blackbox filters (L5).
5. Add automated tests where there are none: imports (I1–I3, I5, I7, I8), SDKs (K3, K4), web UI (L*).
6. Run the VM integration suites in CI (M2).

A Install & setup

7 of 9 done

ID	Feature · sub-features	Tested by
 A1	One-line installer (macOS / Linux) ✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2	Windows installer (install.ps1) ✓ Installs WSL, elevates, resumes after reboot ✓ Downloads vdb.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>vdb setup</code>	Pester (Windows CI); Windows run 16 Sep
 A3	vdb setup on macOS (Lima) ✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack	Manual (from-scratch VM run)
 A4	vdb setup on Windows (WSL2 + btrfs) ✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5	Linux native ✓ Engine runs in-process, no VM	VM integration (all suites run on Linux)
 A6	Command forwarding & host file streaming ✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro	Unit (<code>TestImportLocalFile</code> , <code>TestWSLArgs</code>)
 A7	Always-latest engine binary ✓ Downloads the newest engine into the cache (ELF check) ✓ <code>VECTORADB_NO_REFRESH</code> , <code>VDB_VERSION</code>	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8	New-release notice on vdb start ✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release	Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check
 A9	vdb update ✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>vdb</code> , keeps previous binaries	Unit; update \$4–\$7; a real Mac run

B CLI & services

1 of 2 done

ID	Feature · sub-features	Tested by
 B1	The vdb command set ✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites

✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .vectoradb</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1
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C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓ C1	Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	VM §2 (ZFS); btrfs manual on Windows
✓ C2	Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume	Unit; VM §2, §4
✓ C3	Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓ C4	Private networking (no host ports) ✓ Branch databases reachable only through the gateway	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>)
✓ C5	Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🕒 C6	Storage quotas ✓ Space reserved for main (ZFS) 🕒 Per-branch cap (opt-in)	Manual
✓ C7	Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓ C8	Provisioning & image pull ✓ Docker, storage pool and image prepared on up	Manual

D Durability

2 of 4 done

ID	Feature · sub-features	Tested by
✓ D1	WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path	VM §3
✓ D2	Base backups ✓ Create and list □ Schedule and retention	VM §3
🕒 D3	Point-in-time restore ✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials	VM §3; v2 §1a
🕒 D4	HA standby & failover ✓ Streaming hot standby ✓ Manual failover, main rerouted ✓ Nothing deletes or stops the promoted standby ✓ vdb up brings it back, not the old main	Unit; VM §6, §7

E Security foundations

3 of 3 done

ID	Feature · sub-features	Tested by
 E1	Per-install generated secrets ✓ Generated Postgres and MinIO passwords ✓ Used by restore too	Unit (indirect); v2 §1a
 E2	TLS everywhere ✓ Self-signed TLS on all three services, zero setup 🟡 TLS optional at the gateway (<code>sslmode=disable</code> still connects)	Unit (<code>TestReadStartupUpgradesTLS</code>)
 E3	Zero-signup local API key ✓ First start creates a user, key and connection string 🟡 Windows setup summary doesn't show it	Manual

F SQL gateway

7 of 7 done

ID	Feature · sub-features	Tested by
 F1	Routing by database name ✓ One endpoint :6432, database name = branch ☐ Connection pooling, rate limiting	Unit; VM §1, §6
 F2	API-key authentication ✓ <code>vdb_</code> key as the Postgres password ✓ Bad key rejected with a Postgres error	VM §1
 F3	Backend login (SCRAM) ✓ Cleartext, md5, SCRAM-SHA-256 ✓ Clients never see the database password ☐ Unit test for SCRAM	VM §1, §6, §8 (exercised)
 F4	Per-user Postgres login roles ✓ Role per user, actor can't be forged ✓ Permissions of non-superuser <code>vdbclient</code>	Manual
 F5	Attribution session settings ✓ Actor kind, branch, session id per connection 🟡 Kind / branch / session client-settable	VM §8
 F6	Ctrl-C (CancelRequest) forwarding ✓ Cancel a running query through the gateway	Unit
 F7	Errors & insecure build tag ✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (<code>TestSendError</code>)

G Accounts & auth

4 of 5 done

ID	Feature · sub-features	Tested by
 G1	Auth store (SQLite) ✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once ☐ Roles / permissions, multi-tenant projects	Unit (<code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code>); VM §1
 G2	Email & password users ✓ Create account, log in ☐ Password change / reset ☐ List and delete users	Unit; VM §1
 G3	API keys ✓ Create (shown once), list, revoke ✓ Bearer / X-API-Key and gateway password 🟡 Revoking an unknown id reports success	Unit; VM §1
 G4	Web sessions & signup control ✓ <code>httpOnly</code> session cookie ✓ Signup can be closed	Unit (<code>TestSessions</code>)
 G5	OAuth (GitHub / Google) ✓ Sign-in flows built ☐ Correct default URLs ☐ Tested with a real OAuth app	None

H Blackbox (record of schema changes)

16 of 18 done

ID	Feature · sub-features	Tested by
✓ H1	DDL recording ✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch	VM §8
🕒 H2	Hash chain, append-only & verify ✓ Rows can't be edited or deleted normally ✓ <code>vdb ledger verify</code> 🕒 Superuser rewrite caught only via anchors (H10–H12)	VM §8; v2 §0, §6b
🕒 H3	Attribution ✓ Humans via their login role ✓ Agents via their own <code>agent-<id></code> role	VM §8; v2 §1c
✓ H4	Destructive-DDL guardrail ✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / <code>vdb_admin</code>, use recorded	VM §8; v2 §1b, §6b
✓ H5	Risk flags ✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals)	Unit; v2 §6b
✓ H6	Blackbox access (CLI / REST / MCP / UI / SDK) ✓ CLI, REST with filters, MCP, web page, SDKs	v2 §0, §5
✓ H7	<code>vdb ledger revert --to</code> ✓ Help and output say it's a point-in-time restore of main	Unit (help text); VM §3 (the restore)
✓ H8	Override permission (<code>vdb_admin</code>) & upgrade ✓ <code>vdb admin grant revoke list</code> ✓ First user granted automatically	v2 §1b, §6
✓ H9	Blackbox 2.0 capture ✓ Transaction id and WAL position per entry ✓ Task, tool-call hash, override use	Unit; v2 §2
✓ H10	Checkpoints & anchors ✓ Merkle root in the database and in read-only anchor files ✓ On a schedule and on demand	Unit; v2 §3, §3b
✓ H11	<code>vdb-verify</code> (independent verifier) ✓ Checks a live branch or an export	Unit; v2 §3
✓ H12	Blackbox integrity & export ✓ INTACT / TAMPERED via CLI, REST, MCP ✓ JSON-lines export	v2 §3
✓ H13	Branch from just before a Blackbox entry ✓ Restore + replay up to the change's transaction ✓ By transaction id or timestamp	Unit; v2 §4
✓ H14	Blackbox names & aliases ✓ <code>vdb blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names	Unit; v2 §5
✓ H15	Policy gate (warn / block) ✓ VDB02 warn / VDB01 block with JSON detail ✓ Blocked attempts recorded ✓ Per-rule <code>vdb_admin</code> override	Unit; v2 §6, §6b

✓	H16 Agent provenance & execute_change	Unit; v2 §7
✓	- Session, task, parent session on every agent change (also through the Gateway DSN) - MCP <code>execute_change</code> with policy preview	✓ <code>vdb_blackbox_sessions</code>, REST ○ Provenance is what the agent says (not verified)
✓	H17 Impact analysis	Unit; v2 §8
✓	- Dependents (views, foreign keys, indexes, depth) - Other branches that have the object - Policy verdict and risk score	✓ CLI, REST, MCP, inside <code>execute_change</code> ○ Functions, types, schemas not recognised from SQL
✓	H18 Blackbox diff between branches	Unit; v2 §8
✓	- Fork point, changes only on each side - Objects changed on both	✓ CLI, REST, MCP □ Merge (diff only today)

I Import & migration 12 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres	Manual
✓	- Schema and data via <code>pg_dump</code> - Zero-table result reported as an error	□ Automated test
✓ I2	From MySQL ≤5.7 / MariaDB	Manual
✓	<code>pgloader</code> with type mapping (multi-arch image)	□ Automated test
✓ I3	From MySQL 8.x	Manual
✓	- Native <code>mysqldump</code> path (<code>caching_sha2_auth</code>) - Types, NULLs, unicode, PK	□ Other indexes, foreign keys, defaults □ Automated test
✓ I4	From MongoDB	VM §9, §10
✓	Collection → typed table, exact name	✓ Nested objects / arrays kept as jsonb
✓ I5	From a .sql dump	Unit; VM §11
✓	- Runs the dump into a new branch - Fails listing every failed statement (instance kept)	✓ Same for Postgres imports (I1) ✓ Automated test
✓ I6	From JSON / NDJSON	Unit; VM §10, §11
✓	Columns inferred from top-level keys	✓ An invalid document fails the import, naming its element or line
✓ I7	From CSV	Manual
✓	Table from the header, exact column names	□ Typed columns (all text today)
○ I8	Continuous replication & cutover	Manual
✓	- Initial copy + streaming (logical replication) - Cutover keeps data (CLI)	□ Cutover from REST / web □ Automated test
✓ I9	Guard handling during import	Manual
✓	Guardrail paused, recording stays on	✓ Re-armed even after a crash
✓ I10	Live progress (CLI & SSE)	Manual
✓	Live log and progress bar	○ <code>psql</code> subprocess output not streamed
✓ I11	Import REST endpoints	Manual
✓	From a connection string or an uploaded file	□ Target name validation
✓ I12	Web Import page	Manual
✓	Database and file panels with live console	○ Mentions a SQLite loader that doesn't exist
✓ I13	Exact identifier fidelity	Unit; VM §10
✓	Case kept (<code>userId</code> , <code>leaseAiChats</code>)	○ Field order of JSON sources not kept

J ETL pipelines (experimental) 4 of 4 done

ID	Feature · sub-features	Tested by
 J1	Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating ☐ Dependency graph ☐ Incremental models ☐ Scheduling	Unit; VM §9
 J2	Data-quality tests ✓ <code>not_null</code>, <code>unique</code>, <code>accepted_values</code>, <code>row_count_min</code>, <code>custom</code> ✓ Failure → exit 1, data kept	Unit; VM §9
 J3	Stored pipelines & run history ✓ Saved per user, run over the API, history ☐ Read a run's log ☐ Mark runs whose process died	Manual
 J4	Pipeline editor (web) ✓ Create, edit, run, results, history ☐ Confirm before delete	Manual

K

APIs, SDKs & agents

5 of 6 done

ID	Feature · sub-features	Tested by
 K1	Control-plane REST API ✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys ☐ Versioned paths (<code>/api/v1</code>)	Unit; VM §1; v2 §0–§8
 K2	OpenAPI spec ✓ Served, Blackbox 2.0 endpoints documented ✓ Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases) ✓ Drift check in CI	Unit (drift check)
 K3	Python SDK ✓ Branches, query, Blackbox ☐ Import, pipelines, keys, agents ☐ Tests ☐ Publish to PyPI	Manual
 K4	TypeScript SDK ✓ Same client for Node and browsers ☐ Tests ☐ Publish to npm	Manual
 K5	Agent Branch API ✓ A branch per agent, own non-superuser role ✓ Provenance (task, parent, session) ✓ DSN reachable from the host (gateway-routed, branch-scoped key) ✓ Cap, listing and TTL count suspended branches	Unit; VM §5; v2 §1c, §7
 K6	MCP server ✓ Branches, <code>run_sql</code>, Blackbox tools ✓ <code>policy_check</code>, <code>execute_change</code>, <code>impact</code>, <code>diff</code> ☐ Authentication ✓ Client config & 16-tool reference (<code>docs/mcp.md</code>)	Unit; v2 §1c–§8

L

Web console

9 of 10 done

ID	Feature · sub-features	Tested by
 L1	Embedded single-origin UI ✓ Served by the engine at <code>:8080</code> ☐ Automated UI tests (whole area)	Manual
 L2	Login & signup ✓ Log in / sign up tabs, OAuth buttons	Manual
 L3	Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN	Manual
 L4	SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks	Manual; API in v2 §6

	L5 Blackbox page ✓ Entries per branch with badges □ Table / risk / date filters	□ Verify button, statement and session	Manual
	L6 API keys page ✓ Create, copy, list, revoke		Manual
	L7 Landing, Guide & Docs pages ✓ Landing, developer guide, reference	✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
	L8 Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI	🟡 Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
	L9 Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch	✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep
	L10 App sidebar layout ✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists	✓ No console errors on any page □ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep

M

Build, CI & releases

6 of 6 done

ID	Feature · sub-features	Tested by
 M1	Makefile targets ✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded) ✓ release, web, WSL builds, feature-doc (both PDFs)	CI
 M2	Linux CI ✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check ❑ VM integration suites in CI	CI
 M3	Windows CI ✓ Build, vet, test, Pester, installer encoding ❑ End-to-end WSL install test	CI
 M4	Releases & GHCR image ✓ Binaries for 5 platforms + vdb-verify ✓ SHA256SUMS merged safely between jobs ✓ Fails unless vdb update 's files are published ✓ Multi-arch Postgres image on GHCR	Script tests in CI
 M5	WSL distro & ZFS builds ✓ Prebuilt distro image ✓ Downloaded and verified by the installer (seen on Windows) ✓ Preloaded image tagged as the engine runs it ✓ Imported on a fresh Windows install; preloaded images loaded 🟡 685 MB download (rootfs fallback covers failures)	Actions artifacts; Pester; Windows run 16 Sep
 M6	Postgres image ✓ Postgres 16 + wal-g ❑ pgvector	Manual

N Not started (roadmap)

11 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐ -	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐ -	Branch merge ☐ Apply one branch's schema changes to another	H18
☐ -	Scheduled backups & retention	D2
☐ -	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐ -	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐ -	More import sources: Cassandra, DynamoDB	I1–I8
☐ -	pgvector in the Postgres image	M6
☐ -	Cloud write-once anchors (S3 Object Lock etc.)	H10
☐ -	Release signatures	A9, M4
☐ -	Hosted cloud tier	—
☐ -	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.